Lecture 29

Numerical and signed series

Uses.

Summary 3: Sequence limits obey arithmetic laws. A convergent sequence is bounded, and $q^n \to 0$ for $|q| < 1$.
Summary 4: A bounded monotone real sequence converges. Bolzano–Weierstrass gives a convergent subsequence of any bounded real sequence. Squeeze and subsequence limits hold; $c^{1/n} \to 1$ for $c > 0$ and $n^{1/n} \to 1$.
Summary 25: Bounded monotone functions on finite closed intervals are integrable. Integral order and additivity hold.
Summary 26: Primitives evaluate continuous finite integrals.
Summary 27: Nonnegative improper integrals converge exactly when their cutoff values are bounded; tail power integrals converge for $p > 1$.

Objectives. *An infinite sum is a limit of finite partial sums. We first establish convergence tests for nonnegative terms, then handle signs by absolute convergence and alternating estimates. The tests also prove two growth limits needed for power and Taylor series.*

By the end of this lecture you will be able to:

- Define series convergence and distinguish it from convergence of the terms to zero.
- Apply positive comparison, ratio, root and integral tests with their exceptional cases.
- Prove absolute convergence implies convergence and derive the alternating error bound.
- Bound a convergent series tail and establish the stated factorial and geometric growth limits.

29.1 Partial sums define the infinite sum

Definition 29.1. For a real sequence $(a_n)_{n\geq 1}$, its *partial sums* are $S_N = \sum_{n=1}^{N} a_n$. The *series* $\sum_{n=1}^{\infty} a_n$ *converges* to $S \in \mathbb{R}$ if $S_N \to S$. Otherwise it *diverges*. The number S, when it exists, is its sum.

Adding or removing finitely many initial terms changes partial sums by a fixed finite amount once the cutoff passes those terms. Thus it changes the sum accordingly, without changing convergence. Series starting at index 0 use the same definition with that initial index. Linearity follows from finite-sum linearity and the limit laws: if two series converge, their termwise linear combination converges to the same linear combination of their sums.

Convergence forces $a_n \to 0$. Indeed, $a_n = S_n - S_{n-1}$ for $n \geq 2$, and both partial-sum sequences tend to S, so their difference tends to zero. This *term test* disproves convergence

when terms do not tend to zero. A zero term limit by itself proves nothing about the sum.

Example 29.2 (Geometric and telescoping sums). For $q \in \mathbb{R}$ with $q \neq 1$, multiply the finite sum $1 + q + \cdots + q^N$ by $1 - q$. All middle terms cancel, giving

$$\sum_{n=0}^{N} q^n = \frac{1 - q^{N+1}}{1 - q}.$$

If $|q| < 1$, Lecture 3 gives $q^{N+1} \to 0$, so $\sum_{n=0}^{\infty} q^n = 1/(1-q)$. If $|q| \geq 1$, the terms do not tend to zero and the series diverges, including $q = 1$ and $q = -1$.

For a second sum, the identity $1/(n(n+1)) = 1/n - 1/(n+1)$ gives

$$\sum_{n=1}^{N} \frac{1}{n(n+1)} = 1 - \frac{1}{N+1} \longrightarrow 1.$$

This is a *telescoping series*: cancellation in the finite sum leaves only boundary terms. The cancellation was done before taking its limit.

The harmonic series $\sum_{n=1}^{\infty} 1/n$ supplies the term-test guard. Its terms tend to zero, but for every integer $m \geq 1$,

$$S_{2^m} = 1 + \sum_{j=1}^{m} \sum_{n=2^{j-1}+1}^{2^j} \frac{1}{n} \geq 1 + \sum_{j=1}^{m} \frac{2^{j-1}}{2^j} = 1 + \frac{m}{2}.$$

Each block contains 2^{j-1} terms, each at least $1/2^j$. The displayed subsequence is unbounded, so the series diverges.

29.2 A criterion in terms of finite tails

Convergence can be tested without knowing the sum beforehand. The next criterion says that every sufficiently late finite tail must be small, regardless of how many terms that tail contains.

Theorem 29.3 (Cauchy criterion for a series). *A real series converges if and only if, for every $\eta > 0$, some integer N satisfies*

$$\left| \sum_{k=n+1}^{m} a_k \right| < \eta \qquad \text{for all integers } m > n \geq N.$$

Proof. If $S_n \to S$, choose N such that $|S_n - S| < \eta/2$ for $n \geq N$. The tail is $S_m - S_n$, whose absolute value is less than η by the triangle inequality. This proves necessity.

Conversely, assume the tail condition. With tolerance 1, it bounds all late partial sums within 1 of one fixed partial sum. The finitely many earlier sums are bounded as well, so (S_n) is bounded. Bolzano–Weierstrass from Lecture 4 gives a subsequence $S_{n_j} \to S$. For a new tolerance $\eta > 0$, apply the tail condition with $\eta/2$ and choose one subsequence index $n_j \geq N$ with $|S_{n_j} - S| < \eta/2$. For every $n \geq N$, the tail condition bounds $|S_n - S_{n_j}| < \eta/2$ when the indices differ; if equal, this difference is zero. It applies in either index order because of the absolute value. Thus $|S_n - S| < \eta$ by the triangle inequality. All partial sums converge to S. ∎

The term test concerns a single late term. This criterion controls an entire late block, with no fixed bound on the block's length. Harmonic terms tend to zero, but their dyadic blocks in the preceding example are at least $1/2$, violating the criterion.

29.3 Nonnegative comparison

If $a_n \geq 0$, then $S_{N+1} - S_N = a_{N+1} \geq 0$. The monotone sequence theorem from Lecture 4 therefore gives a criterion: *a nonnegative series converges exactly when its partial sums are bounded above*. If they are unbounded, they increase to $+\infty$.

Theorem 29.4 (Comparison and limit comparison). *Suppose $0 \leq a_n \leq b_n$ for all sufficiently large n. Convergence of $\sum b_n$ implies convergence of $\sum a_n$; divergence of $\sum a_n$ implies divergence of $\sum b_n$. If $a_n, b_n > 0$ eventually and $a_n/b_n \to L \in (0, \infty)$, the two series either both converge or both diverge.*

Proof. Remove the finitely many terms before the inequalities hold. Every partial sum of a_n is bounded above by the corresponding partial sum of b_n. If the latter converge, they are bounded, so the former are bounded and converge by the nonnegative criterion. The divergence statement is the contrapositive. For limit comparison, eventually $(L/2)b_n \leq a_n \leq (3L/2)b_n$. Apply comparison in both directions and use that a positive constant multiple does not change convergence. ∎

Example 29.5 (A geometric majorant). Test $\sum_{n=1}^{\infty} 1/(3^n + n)$. Each term is positive and at most 3^{-n}. The geometric series $\sum_{n=1}^{\infty} 3^{-n}$ converges, so comparison proves convergence. The estimate establishes existence of the sum without claiming an exact value.

29.4 Absolute convergence and the ratio test

Definition 29.6. A real series *converges absolutely* if $\sum |a_n|$ converges. It *converges conditionally* if $\sum a_n$ converges while $\sum |a_n|$ diverges.

Theorem 29.7 (Absolute convergence implies convergence). *Every absolutely convergent real series converges.*

Proof. Define $a_n^+ = (|a_n| + a_n)/2$ and $a_n^- = (|a_n| - a_n)/2$. Both lie between zero and $|a_n|$, so comparison gives convergent series of positive and negative parts. Their partial sums have finite limits. Since $a_n = a_n^+ - a_n^-$, the difference limit law gives convergence of $\sum a_n$. This argument never subtracts infinite quantities. ∎

The same finite triangle inequality and limit passage give $|\sum a_n| \leq \sum |a_n|$ whenever the right side converges.

Theorem 29.8 (Ratio test). *Suppose $a_n \neq 0$ eventually and $|a_{n+1}|/|a_n| \to \ell$, where $0 \leq \ell \leq \infty$. If $\ell < 1$, the series converges absolutely. If $\ell > 1$, it diverges. At $\ell = 1$ the test gives no conclusion.*

Proof. If $\ell < 1$, choose r with $\ell < r < 1$. From some index N onward, $|a_{n+1}| \leq r|a_n|$. Induction gives $|a_{N+j}| \leq |a_N| r^j$ for every integer $j \geq 0$. The geometric majorant converges, so comparison and Theorem 29.7 give absolute convergence. If $\ell > 1$, choose $r > 1$ smaller than

ℓ; any fixed $r > 1$ works eventually when $\ell = \infty$. The reverse bound gives $|a_{N+j}| \geq |a_N| r^j$, with $|a_N| > 0$. These terms do not tend to zero, so the term test gives divergence. ∎

The nonzero hypothesis is needed to form the ratios. A series with only finitely many nonzero terms converges directly, even though this ratio formula is eventually undefined.

Example 29.9 (Factorials and two growth limits). For $a_n = 3^n/n!$, the ratio is $3/(n+1) \to 0$, so its series converges. Its sum will be identified in Lecture 31. More generally, for any fixed $c \in \mathbb{R} \setminus \{0\}$, the positive terms $|c|^n/n!$ have ratio $|c|/(n+1) \to 0$. The convergent-series term test and $|c^n/n!| = |c|^n/n!$ give

$$\frac{c^n}{n!} \longrightarrow 0 \quad (c \in \mathbb{R}), \qquad nq^n \longrightarrow 0 \quad (|q| < 1). \tag{29.1}$$

The first limit includes $c = 0$ directly for $n \geq 1$. For the second, if $0 < |q| < 1$, the terms $n|q|^n$ have ratio $((n+1)/n)|q| \to |q| < 1$. Their series converges by the ratio test, so the terms tend to zero. The signed terms nq^n have those absolute values and also tend to zero. For $q = 0$, they are zero for every $n \geq 1$. These are proved growth limits, available before the power-series arguments.

The geometric comparison also bounds the truncation error. Fix an integer $N \geq 1$ and suppose $0 \leq r < 1$ and $|a_{n+1}| \leq r|a_n|$ for every $n \geq N+1$. Induction bounds the absolute terms after index N by $|a_{N+1}|, |a_{N+1}|r, |a_{N+1}|r^2, \ldots$. The geometric majorant first proves absolute convergence of the series. Summing that majorant and using the absolute-tail triangle inequality gives

$$\left| \sum_{n=1}^{\infty} a_n - \sum_{n=1}^{N} a_n \right| \leq \frac{|a_{N+1}|}{1-r}. \tag{29.2}$$

The first omitted term, rather than the last retained term, starts this bound. For a series starting at index 0, add the same initial term a_0 to the full sum and the partial sum. Their difference, and hence this bound, is unchanged. The inequality itself suffices; unlike a ratio limit, it need not divide by a nonzero term. For $r = 0$ all terms after the first omitted one vanish.

Example 29.10 (A certified factorial-sum approximation). Approximate $S = \sum_{n=0}^{\infty} 3^n/n!$ with error less than $1/1000$. The ratio test already proved convergence. Retain indices 0 through 12. For every term index $n \geq 13$, the next-term ratio is $3/(n+1) \leq 3/14$. Thus the first omitted term has index 13 and Eq. (29.2) gives

$$0 < S - S_{12} \leq \frac{3^{13}}{13!} \frac{1}{1 - 3/14} = \frac{6561}{20134400} < \frac{1}{1000}.$$

Finite rational addition gives $S_{12} = 39591971/1971200$. Together these exact bounds imply $20.085 < S < 20.086$. No identification of S with an elementary function was required. A ratio test can therefore supply both convergence and a numerical error certificate, once an eventual ratio bound has been made explicit.

29.5 The root and integral tests

Theorem 29.11 (Root test; proof on Sheet 29). *If $|a_n|^{1/n} \to \ell \in [0, \infty]$, then $\sum a_n$ converges absolutely for $\ell < 1$ and diverges for $\ell > 1$. At $\ell = 1$ the test gives no conclusion.*

Sheet 29 supplies the full proof by geometric comparison. Unlike the ratio test, the root test allows zero terms without any undefined quotient. The root limits in Theorem 4.4 give $n^{1/n} \to 1$. For instance, the terms $a_n = n^2(2/5)^n$ have $|a_n|^{1/n} = (n^{1/n})^2(2/5) \to 2/5$, so their series converges. For $1/n$ and $1/n^2$, both root and ratio limits are 1. The harmonic example diverges; the next theorem proves that the square-power example converges. Neither test decides the borderline by itself.

Figure 29.1 shows the finite rectangle bounds behind the integral test. The hypothesis is monotonicity of a nonnegative function.

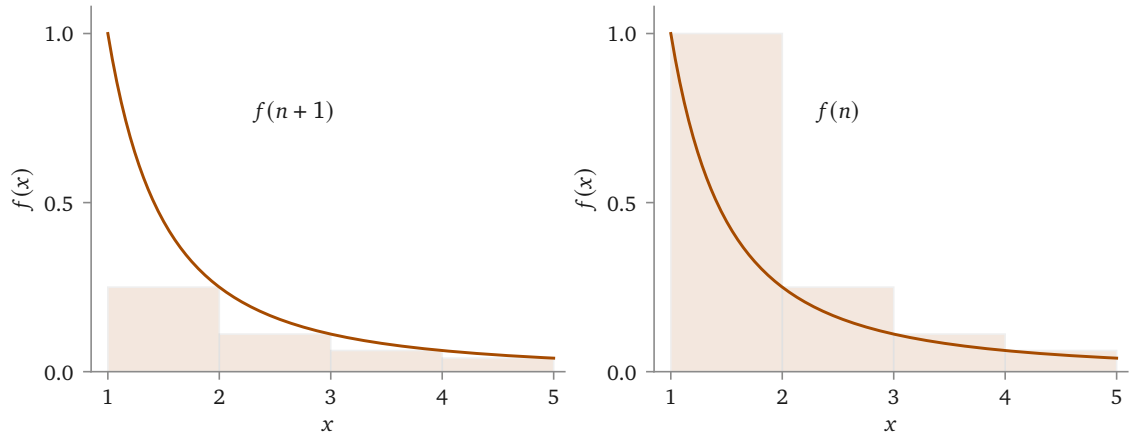


Figure 29.1: For a nonincreasing function, right-endpoint rectangles lie below the graph and left-endpoint rectangles above it. Summing their areas compares a finite sum with a finite integral.

Theorem 29.12 (Integral test and remainder bounds). *Let $f : [1, \infty) \to [0, \infty)$ be nonincreasing. Then $\sum_{n=1}^{\infty} f(n)$ converges if and only if $\int_1^{\infty} f(x)\,dx$ converges. When they converge, the remainder after index $N \geq 1$ satisfies*

$$\int_{N+1}^{\infty} f(x)\,dx \leq \sum_{n=N+1}^{\infty} f(n) \leq \int_{N}^{\infty} f(x)\,dx.$$

Proof. A nonincreasing real function is bounded and Riemann integrable on every finite interval here, by Lecture 25. On $[n, n+1]$ its values lie between $f(n+1)$ and $f(n)$. Integral order and additivity give

$$\int_1^{N+1} f \leq \sum_{n=1}^{N} f(n) \leq f(1) + \int_1^{N} f.$$

If the improper integral converges, its nonnegative cutoff values are bounded, hence so are the series partial sums. Conversely, bounded partial sums bound the integral at integer cutoffs. At any real cutoff, bound it by the integral at a larger integer using positivity. The nonnegative cutoff criterion of Lecture 27 proves convergence of the integral. For the remainder, the same rectangle bounds give, for $M > N$,

$$\int_{N+1}^{M+1} f \leq \sum_{n=N+1}^{M} f(n) \leq \int_{N}^{M} f.$$

Every displayed limit exists in the convergent case. Letting $M \to \infty$ proves the bounds. ∎

For $p > 0$, the function x^{-p} is positive and nonincreasing. The improper power test from Lecture 27 therefore shows

$$\sum_{n=1}^{\infty} n^{-p} \text{ converges exactly when } p > 1.$$

For $p \le 0$, the terms are at least 1 and divergence follows from the term test. Thus the statement holds for every real p.

Example 29.13 (An interval for a sum). Bound $S = \sum_{n=1}^{\infty} n^{-3}$ using its first four terms. This series converges by the power test just proved. Its partial sum is $S_4 = 1 + 1/8 + 1/27 + 1/64 = 2035/1728$. The remainder bounds integrate x^{-3}, giving

$$\frac{2035}{1728} + \frac{1}{50} \le S \le \frac{2035}{1728} + \frac{1}{32}.$$

The interval is rigorous without an exact closed formula for the sum. Choosing more terms narrows both endpoint bounds by the same proved method.

29.6 Alternating sums and their error

Theorem 29.14 (Alternating-series test). *If $b_n \ge 0$, $b_{n+1} \le b_n$ and $b_n \to 0$, then $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ converges to a number S. For every $N \ge 1$, its partial sum A_N satisfies*

$$0 \le (-1)^N (S - A_N) \le b_{N+1}.$$

Proof. The even sums increase, since $A_{2m+2} - A_{2m} = b_{2m+1} - b_{2m+2} \ge 0$. The odd sums decrease, since $A_{2m+1} - A_{2m-1} = -b_{2m} + b_{2m+1} \le 0$. Also $A_{2m} \le A_{2m+1}$, and all these sums lie between 0 and b_1: pairing adjacent terms gives the lower bound for even sums, and pairing terms after b_1 gives the upper bound for odd sums. Both subsequences therefore converge by monotone convergence. Their difference $A_{2m+1} - A_{2m} = b_{2m+1} \to 0$ makes their limits equal. Every partial sum is even or odd, so the whole sequence converges to that common value S.

Monotonicity places S above every even partial sum and below every odd partial sum. For either parity of N, the sum thus lies between A_N and A_{N+1}. Their difference is $(-1)^N b_{N+1}$. This proves both the sign and magnitude of the stated remainder bound. ∎

The alternating harmonic series $\sum_{n=1}^{\infty} (-1)^{n-1}/n$ converges by this theorem, but its absolute-value series is the divergent harmonic series. It is therefore conditionally convergent. After 100 terms its error is at most $1/101 < 1/100$. Lecture 31 will identify its sum; that value is not needed for convergence or error control.

Grouping consecutive terms is a separate operation from changing their order. Let $0 = N_0 < N_1 < N_2 < \cdots$ be integers, and replace each finite block by $c_j = \sum_{n=N_{j-1}+1}^{N_j} a_n$. The grouped partial sums are exactly S_{N_j}. Since $N_j \ge j$, they form a subsequence tending to the original sum whenever the original series converges. Thus such grouping preserves an already established sum.

The converse need not hold. Grouping $1 - 1 + 1 - 1 + \cdots$ into adjacent pairs gives a series of zeros, but the original partial sums alternate between 1 and 0. The grouped sums have discarded those odd cutoffs. Consequently a convergent grouped series alone does not establish convergence of its ungrouped source. This distinction will matter when separating terms of Taylor series.

A *rearrangement* of a series replaces its order by a bijection $\sigma : \mathbb{N} \to \mathbb{N}$, producing $\sum_{n=1}^{\infty} a_{\sigma(n)}$. For conditionally convergent real series, such a change can alter the sum. Sheet 29 proves this warning and the stronger statement that any prescribed real sum can be obtained. The rearrangement result does not license arbitrary regrouping before convergence has been justified.

Summary 29.

- Series convergence means convergence of partial sums. It forces $a_n \to 0$; the harmonic series shows the converse is false.
- Theorem 29.3: convergence is equivalent to every sufficiently late finite tail being small, uniformly over both tail endpoints.
- Nonnegative series converge exactly when their partial sums are bounded. Comparison and positive finite limit comparison transfer convergence.
- Theorems 29.7, 29.8 and 29.11: absolute convergence implies convergence. Ratio and root limits below 1 give absolute convergence, above 1 give divergence; at 1 they are inconclusive. The root proof is on Sheet 29.
- Theorem 29.12: nonnegative nonincreasing functions give an integral test with two-sided tail bounds. The series $\sum n^{-p}$ converges exactly for $p > 1$.
- Theorem 29.14: nonincreasing nonnegative terms tending to zero give an alternating sum with error bounded by the first omitted term.
- An eventual geometric bound on consecutive absolute terms gives the quantitative remainder estimate Eq. (29.2), starting with the first omitted term.
- The limits in Eq. (29.1) hold for every real c and $|q| < 1$. Conditional rearrangements can change a sum; the proof is on Sheet 29.

Lecture 30 studies series of functions, where control at each point must be distinguished from control over the whole interval.

References

J. Stewart, D. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th ed., Cengage, 2020, §§11.2–11.7.